02 - Image Preprocessing Pipeline

Milestone 2: Preparing the Rakuten product images for reproducible image and multimodal modelling.

This notebook converts the image-audit findings into a reusable preprocessing workflow. It does not train an image model. Raw files remain unchanged, and images are opened only when they are needed.

1. Scope and Audit Context

The data-audit notebooks established the following points:

- all 84,916 labelled training products have a matching image file;
- all 2,000 sampled images were readable, RGB images with a size of 500 × 500 pixels;
- exact file duplicates were audited for the labelled training catalogue only;
- the portable training-image file hashes were cached under `data/processed/`.

The purpose of this notebook is therefore not to repeat the image audit. It prepares the image manifest, loads the shared validation split created by `02_preprocessing.ipynb`, uses the cached audit metadata to create clean image training and validation sets, and defines deterministic framework-neutral image preprocessing.

2. Imports and Configuration

Project paths and image filenames remain centralised in `src.data_utils`. No dataset path is duplicated in this notebook.

```
Environment and data utilities loaded successfully.
```

3. Build the Training Image Manifest

`load_merged_data()` returns the labelled challenge data. The original dataframe index is retained as `row_id`; this is the safest key for joining text predictions, image predictions and split assignments later in the project.

The manifest stores metadata and file paths only. It does not load 84,916 images into memory.

```
Training image rows: 84,916
```

4. Validate the Manifest

The earlier audit already checked the complete training filename coverage. These assertions protect the preprocessing workflow against accidental path, join or data changes.

```
Available training images: 84,916/84,916
Missing training images  : 0
```

5. Load the Shared Training/Validation Split

The official challenge test data has no labels and is not used for local evaluation. This notebook therefore loads the shared training/validation assignment created from the labelled catalogue in `02_preprocessing.ipynb`. The assignment remains unchanged here so text and image predictions can still be joined through `row_id`.

```
Loaded shared split manifest.
```

split	products	classes	share
train	67932	27	80.0%
validation	16984	27	20.0%

6. Create Duplicate- and Conflict-Free Image Sets

The cached file SHA-256 hashes identify exact image matches. Cleaning is image-specific and does not overwrite the shared split. The following deterministic rules are applied:

1. remove every file-hash group containing more than one `prdtypecode`;
2. keep only one row per remaining file hash;
3. when a hash occurs in both shared partitions, keep one validation row and remove its training copies;
4. retain `row_id` so text predictions can later be restricted to the same clean validation products.

The raw data and the shared `data_split.csv` remain unchanged.

	before	after	classes	unique_file_hashes	removed	removed_share
train	67932	62312	27	62312	5620	8.27%
validation	16984	16433	27	16433	551	3.24%

prdtypecode	train_count	val_count	train_share	val_share	share_diff
10	2483	621	3.98%	3.78%	0.21%
40	1978	494	3.17%	3.01%	0.17%
50	1272	329	2.04%	2.00%	0.04%
60	623	162	1.00%	0.99%	0.01%
1140	2127	532	3.41%	3.24%	0.18%
1160	3135	789	5.03%	4.80%	0.23%
1180	605	153	0.97%	0.93%	0.04%
1280	3528	909	5.66%	5.53%	0.13%
1281	1601	401	2.57%	2.44%	0.13%
1300	3110	915	4.99%	5.57%	0.58%
1301	590	156	0.95%	0.95%	0.00%
1302	1928	493	3.09%	3.00%	0.09%
1320	2556	646	4.10%	3.93%	0.17%
1560	3659	978	5.87%	5.95%	0.08%
1920	3304	855	5.30%	5.20%	0.10%
1940	632	160	1.01%	0.97%	0.04%
2060	3836	985	6.16%	5.99%	0.16%
2220	643	164	1.03%	1.00%	0.03%
2280	3808	952	6.11%	5.79%	0.32%
2403	3817	955	6.13%	5.81%	0.31%
2462	1133	283	1.82%	1.72%	0.10%
2522	3667	971	5.88%	5.91%	0.02%
2582	1895	494	3.04%	3.01%	0.04%
2583	5783	1844	9.28%	11.22%	1.94%
2585	1699	466	2.73%	2.84%	0.11%
2705	2208	552	3.54%	3.36%	0.18%
2905	692	174	1.11%	1.06%	0.05%

```
Excluded conflict rows: 1,074
Excluded rows in total: 6,171
```

7. Export the Clean Image Split

The model-facing manifest contains identifiers, labels, split assignments and file hashes only. Absolute image paths are not stored, so each machine can reconstruct them with `get_image_path()`.

```
Rows: 78,745
```